Writing Out $\mathcal{M}_t^*$ for Compton Scattering

PHYS 7502

Starting Point: $\mathcal{M}_t$

The t-channel amplitude (Griffiths Eq. 7.112):

$$\mathcal{M}_t = \frac{-e^2}{t-m^2}\,\bar{u}(p_4)\,\underbrace{\varepsilon_{3\mu}\gamma^\mu}_{\not{\varepsilon}_3}\,(\not{p}_1 - \not{p}_2 + m)\,\underbrace{\varepsilon^*_{2\nu}\gamma^\nu}_{\not{\varepsilon}^*_2}\,u(p_1) \tag{1}$$

Written out with explicit indices:

$$\mathcal{M}_t = \frac{-e^2}{t-m^2}\,\varepsilon_{3\mu}\,\varepsilon^*_{2\nu}\,\bar{u}(p_4)\,\gamma^\mu(\not{p}_1 - \not{p}_2 + m)\gamma^\nu\,u(p_1) \tag{2}$$

Taking the Complex Conjugate

The Rule (Griffiths Eqs. 7.117–7.119)

For any spinor bilinear:

$$\Big[\bar{u}(a)\,\Gamma\,u(b)\Big]^* = \bar{u}(b)\,\bar{\Gamma}\,u(a)$$

where $\bar{\Gamma} \equiv \gamma^0\Gamma^\dagger\gamma^0$ **reverses the order** of the γ-matrices.

In our case, $\Gamma = \gamma^\mu(\not{p}_1 - \not{p}_2 + m)\gamma^\nu$ with $a = p_4$, $b = p_1$. The bar operation reverses:

$$\bar{\Gamma} = \gamma^\nu(\not{p}_1 - \not{p}_2 + m)\gamma^\mu$$

(The momenta p_1, p_2 are real, so they are unchanged. Only the order flips.)

Also, the polarization vectors pick up conjugation: $\varepsilon_{3\mu} \to \varepsilon^*_{3\mu}$ and $\varepsilon^*_{2\nu} \to \varepsilon_{2\nu}$. The denominator $t - m^2$ is real.

Putting it together:

$\mathcal{M}_t^*$ Written Out

$$\boxed{\mathcal{M}_t^* = \frac{-e^2}{t-m^2}\,\varepsilon^*_{3\mu}\,\varepsilon_{2\nu}\,\bar{u}(p_1)\,\gamma^\nu(\not{p}_1 - \not{p}_2 + m)\gamma^\mu\,u(p_4)} \tag{3}$$

Side-by-Side Comparison

$$\mathcal{M}_t = \frac{-e^2}{t-m^2}\,\varepsilon_{3\mu}\,\varepsilon^*_{2\nu}\,\bar{u}(p_4)\,\gamma^\mu\,(\not{p}_1 - \not{p}_2 + m)\,\gamma^\nu\,u(p_1)$$

$$\mathcal{M}^*_t = \frac{-e^2}{t-m^2}\,\varepsilon^*_{3\mu}\,\varepsilon_{2\nu}\,\bar{u}(p_1)\,\gamma^\nu\,(\not{p}_1 - \not{p}_2 + m)\,\gamma^\mu\,u(p_4)$$

Three things changed:

1. **Spinors swap:** $\bar{u}(p_4)\cdots u(p_1)$ became $\bar{u}(p_1)\cdots u(p_4)$.
2. **γ-matrices reverse:** $\gamma^\mu(\not{p}_1 - \not{p}_2 + m)\gamma^\nu$ became $\gamma^\nu(\not{p}_1 - \not{p}_2 + m)\gamma^\mu$.
3. **Polarizations conjugate:** $\varepsilon_3 \to \varepsilon_3^*$ and $\varepsilon_2^* \to \varepsilon_2$.

Why This Matters: Setting Up Casimir's Trick

When we form $|\mathcal{M}_t|^2 = \mathcal{M}_t \times \mathcal{M}_t^*$, we get:

$$|\mathcal{M}_t|^2 = \frac{e^4}{(t-m^2)^2}\,\varepsilon_{3\mu}\varepsilon^*_{3\alpha}\,\varepsilon^*_{2\nu}\varepsilon_{2\beta}\,\Big[\bar{u}(p_4)\gamma^\mu(\not{p}_1 - \not{p}_2 + m)\gamma^\nu u(p_1)\Big]\Big[\bar{u}(p_1)\gamma^\beta(\not{p}_1 - \not{p}_2 + m)\gamma^\alpha u(p_4)\Big]$$

This has exactly the structure $[\bar{u}(a)\Gamma_1 u(b)][\bar{u}(b)\bar{\Gamma}_2 u(a)]$ that Casimir's trick eats:

$$\sum_{\text{spins}} \to \text{Tr}\Big[\gamma^\mu(\not{p}_1 - \not{p}_2 + m)\gamma^\nu\,(\not{p}_1 + m)\,\gamma^\beta(\not{p}_1 - \not{p}_2 + m)\gamma^\alpha\,(\not{p}_4 + m)\Big]$$

Then the photon polarization sums $\sum \varepsilon_{3\mu}\varepsilon^*_{3\alpha} \to -g_{\mu\alpha}$ and $\sum \varepsilon^*_{2\nu}\varepsilon_{2\beta} \to -g_{\nu\beta}$ contract μ with α and ν with β, producing the final trace shown in the main derivation.